

Territory Briefing: Torino

Evidence-based preliminary infrastructure intelligence for technical meetings, early design discussion and due diligence. ARCUS supports screening; it does not replace site-specific checks or design verification.

REPORT ID

ARCUS-P01-20260603-torino

PROJECT CONTEXT

Railway crossing

DOMINANT DRIVER

Hydraulic

EVENTS

42

SPATIAL LEVEL

Province level

ANALYSIS TYPE

Territory Briefing

VERSION

ARCUS Path 01 v1.0

GENERATED BY

ARCUS Professional

Executive Summary

ARCUS identifies 42 documented collapse precedents in the selected province and translates the Step 3 hazard-overlay state into operational next steps for railway crossing interventions. The dominant territorial signal is **Hydraulic exposure**; the leading historical cause is **Hydraulic**.

Recommended use: province-level preliminary screening before design, due diligence or site-specific investigation.

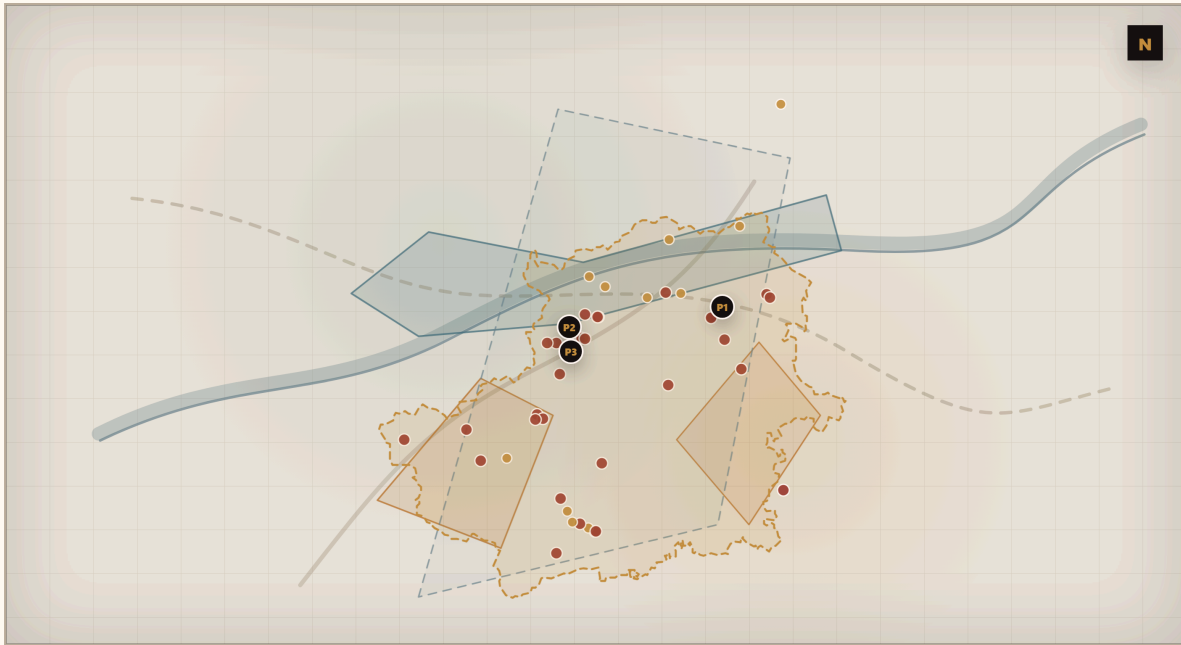
Dominant exposure
**Hydraulic exposure / score
97**

Historical signal
**Hydraulic represents 98% of
the selected evidence.**

Priority use
**Identify railway sections
intersecting flood-prone or
hydraulic-sensitive
corridors.**

02

Selected Province & Real Atlas Map



- **TOTAL COLLAPSE** ● **TRIGGERED EVENT** ● **HYDRAULIC OVERLAY**
- - - **LANDSLIDE OVERLAY** - - - **SEISMIC CONTEXT**

Territorial screening map. Not for design scale. P1/P2/P3 markers identify priority events listed in the table below.

03

How to Read This Report

ARCUS combines historical collapse records, public hazard context, source-backed evidence and comparable precedents. The output is intended for preliminary screening and planning of follow-up checks, not for structural safety certification.

SELECTED PROJECT CONTEXT

Railway crossing

Railway bridge, culvert, embankment or track crossing: service continuity, embankment stability, hydraulic openings, inspections and operational consequences.

DOMINANT EXPOSURE

Hydraulic exposure

41 matched ARCUS events. Public overlays are used as screening context and must be followed by site-specific checks.

04

Key Indicators

ARCUS EVENTS

42

TOTAL
COLLAPSE

30

TRIGGERED

41

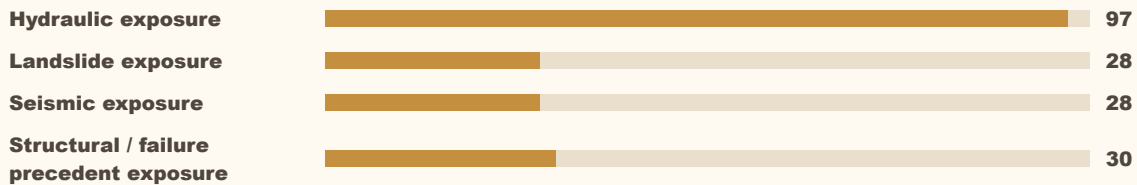
SOURCES

125

05

Hazard / Exposure Context

The Step 3 view combines hydraulic WMS, landslide WMS, seismic context and ARCUS historical collapses. The bars below summarise the province-level exposure signal used by the recommendation engine.



06

Operational Meaning

PRIMARY FOCUS

Track continuity; railway embankment stability; hydraulic openings.

For railway crossing interventions in the selected province, ARCUS indicates that preliminary attention should focus on track continuity and related checks because the dominant indicator is hydraulic exposure. This is a province-level screening signal and must be translated into site-specific checks by qualified professionals.

ATTENTION POINTS

- Track continuity
- Railway embankment stability
- Hydraulic openings
- Scour near supports
- Service interruption consequences
- Access for inspection and emergency response

07

Priority ARCUS Precedents

P1/P2/P3 are shown on the Step 3 map. These records are historical analogues for screening and investigation planning, not direct predictions.

P1**B00.10.15 - Castellamonte 2000-10-13 / Hydraulic****TC / High****P2****B00.10.03 - Ala di Stura 2000-10-13 / Hydraulic****TC / High****P3****B00.10.05 - Ala di Stura 2000-10-13 / Hydraulic****TC / High**

08

Comparable Precedents

Comparable precedents are selected by severity, trigger, dominant driver, evidence confidence and human impact. For this context, the useful question is not whether a future event is predicted, but which mechanisms and data gaps should be checked before design decisions.

09

Interpretation

For railway crossing interventions in Torino, ARCUS indicates that preliminary investigation should be framed around hydraulic exposure, documented collapse precedents and the operational consequences implied by the selected project context.

10

Immediate Screening Actions

ACTION 1

Identify railway sections intersecting flood-prone or hydraulic-sensitive corridors.

ACTION 2

Review hydraulic openings, culverts and bridge supports for preliminary exposure.

ACTION 3

Consider operational continuity and inspection access during hydraulic events.

11

Data to Request Before Design Decisions

DATA 1

Railway alignment

DATA 2

Embankment geometry

DATA 3

Hydraulic opening data

DATA 4

Bridge/culvert inventory

DATA 5

Historical service interruption records

DATA 6

Inspection records

12

Design-Phase Checks

ACTION 1

Translate province-level screening into site-specific checks by qualified professionals.

ACTION 2

Compare planned intervention assumptions with P1/P2/P3 historical precedents.

ACTION 3

Use public WMS overlays as context, not as detailed hydraulic, geotechnical or seismic modelling.

13

Recommended Client Path

1. Identify railway sections intersecting flood-prone or hydraulic-sensitive corridors.
2. Review hydraulic openings, culverts and bridge supports for preliminary exposure.
3. Consider operational continuity and inspection access during hydraulic events.
4. Compare the planned intervention with ARCUS priority precedents around Castellamonte, Ala di Stura, Felleto, Balme, Chialamberto.
5. Request the missing technical data before design or institutional decisions.

14

Data Coverage & Limitations

Province-level screening only; not design scale.

ARCUS does not certify structural safety or asset condition.

Public overlays support prioritisation and must be followed by site-specific investigations.

15

Methodology Snapshot

ARCUS combines documented collapse events, source reliability, spatial relevance, severity, triggers, cause similarity, territorial hazard context and comparable precedents. The dominant exposure indicator is calculated from Step 3 scores: hydraulic, landslide, seismic and structural/failure precedent exposure.

16

Score and Class Legend

P1/P2/P3 identify the top priority historical precedents on the exported Atlas map. Severity TC means total collapse; triggered events indicate an external initiating condition was recorded. Prefer "Failure precedent exposure" unless asset-level structural data are available.

A

Short Source Appendix

B24.09.01 / PROTEZIONE CIVILE

Evento meteo-idrologico 05 settembre 2024

B17.06.01 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B08.05.02 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B08.05.02 / REGIONE PIEMONTE

Evento alluvionale 29-30 maggio 2008 – relazione tecnica ufficiale

B05.11.01 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B00.10.03 / ARPA PIEMONTE

Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale

B00.10.03 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century: survey and statistical analysis

B00.10.04 / ARPA PIEMONTE

Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale

The full source table is provided as a separate supporting CSV export.